BIOLOGY
CLASS 10 / SSC-II

Complete solved subjective papers

Model Paper + 1st Annual Examination 2026
Variants B, D and M | Sections B and C | Total: 53 marks

Paper in the uploaded document	Source pages
Model Question Paper (Curriculum 2022-23)	1-2
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Prepared by Fakhar STEM Sphere
Simple English • All questions and all OR choices • Labelled diagrams • Worked calculations

How to use these answers

• Learn the meaning first, then practise writing the answer in your own simple words. Keep the scientific terms in bold.

• For a 3-mark question, include three relevant points or all named subparts. For a 5-mark answer, use headings, explanation and the requested diagram or calculation.

• Draw diagrams with a pencil, give a title and use straight label lines. The diagrams here are clear study schematics and are not drawn to scale.

• Both sides of every OR are included for preparation. In the exam, answer only the required choice and write the correct question number in the allotted space.

Scope: This is an independently prepared study solution aligned with the questions and marks in your supplied papers. It is not an official FBISE marking scheme; marks depend on the examiner and the accuracy of the student's response. No Section A questions are present in the supplied PDF.

Paper 1: Model Paper

Source: uploaded PDF pages 1-2. Curriculum 2022-23; Scheme of Studies 2006.

Section B: Q.2 (33 marks) - Both choices of every part are answered below for practice. In the examination, follow the choice instructions printed on your paper.

Q.2(i) | Sequence of events during swallowing [3 marks]

• The tongue pushes the chewed food ball, called the bolus, from the mouth into the pharynx.

• The soft palate closes the passage to the nose. The larynx rises and the epiglottis covers its opening, helping prevent food from entering the trachea. Breathing pauses briefly.

• The upper oesophageal sphincter opens. The bolus enters the oesophagus and peristalsis pushes it towards the stomach; the lower sphincter opens to let it in.

Q.2(i) OR | How a kidney transplant helps [3 marks]

A healthy donor kidney is surgically placed in the patient and connected to blood vessels and the urinary bladder through its ureter. It filters the blood, removes urea and extra water and salts, and restores fluid and electrolyte balance. A successful transplant can remove the need for regular dialysis. Tissue matching and anti-rejection medicines reduce the risk of immune rejection.

Q.2(ii) | Why myocardial infarction is life-threatening [3 marks]

A myocardial infarction (heart attack) occurs when a coronary artery becomes blocked. The heart muscle supplied by it receives too little oxygen and nutrients, so some cells die. The damaged heart may pump blood poorly or develop a dangerous irregular rhythm. This can stop oxygen delivery to the brain and other organs and may cause death.

Q.2(ii) OR | Chromosome without a kinetochore [3 marks]

A kinetochore is the protein structure at the centromere where spindle fibres attach. Without it, the chromosome cannot attach properly to the spindle or move normally to a pole. Cell division may stop at a checkpoint; if division continues, the chromosome may be lost or distributed unequally, producing daughter cells with abnormal chromosome numbers.

Q.2(iii) | Inspired and expired air [3 marks]

Component	Inspired air	Expired air
Oxygen	About 21%	About 16%
Carbon dioxide	About 0.04%	About 4%
Nitrogen	About 78%	About 78% (little change)
Water vapour	Variable; usually less	More; nearly saturated
Temperature	Depends on surroundings	Usually warmer, near body temperature

Q.2(iii) OR | Main types of diabetes [3 marks]

Feature	Type 1	Type 2
Cause	Pancreatic beta cells are destroyed; very little or no insulin is made.	Body cells respond poorly to insulin; insulin production may later decrease.
Usual onset	Often young, but possible at any age.	Usually adults, but also occurs in young people.
Control	Insulin, healthy diet and activity.	Diet, exercise and medicines; sometimes insulin.

Q.2(iv) | Components of a reflex arc [3 marks]

Receptor → sensory neuron → relay neuron in the spinal cord → motor neuron → effector (muscle or gland). The receptor detects a stimulus; the effector produces the response. Synapses connect the neurons.

Q.2(iv) OR | Endocrine gland table [3 marks]

Endocrine gland	One hormone	Function
Thyroid	Thyroxine	Regulates metabolic rate and supports normal growth.
Ovary	Oestrogen	Develops female secondary sexual characteristics and helps regulate the menstrual cycle.
Parathyroid	Parathyroid hormone (PTH)	Raises blood calcium when it is too low.

Q.2(v) | Heart wall and septum: X, Y and Z [3 marks]

Labels: X = right ventricular wall; Y = left ventricular wall; Z = interventricular septum.

• A. Why Y is thicker: The left ventricle must pump blood around the whole body at high pressure. The right ventricle pumps only to the nearby lungs at lower pressure, so its wall is thinner.

• B. A pore in Z: A hole allows blood to pass abnormally between the ventricles, usually from left to right because the left pressure is higher. It mixes the circulations, increases blood flow to the lungs and makes the heart work harder.

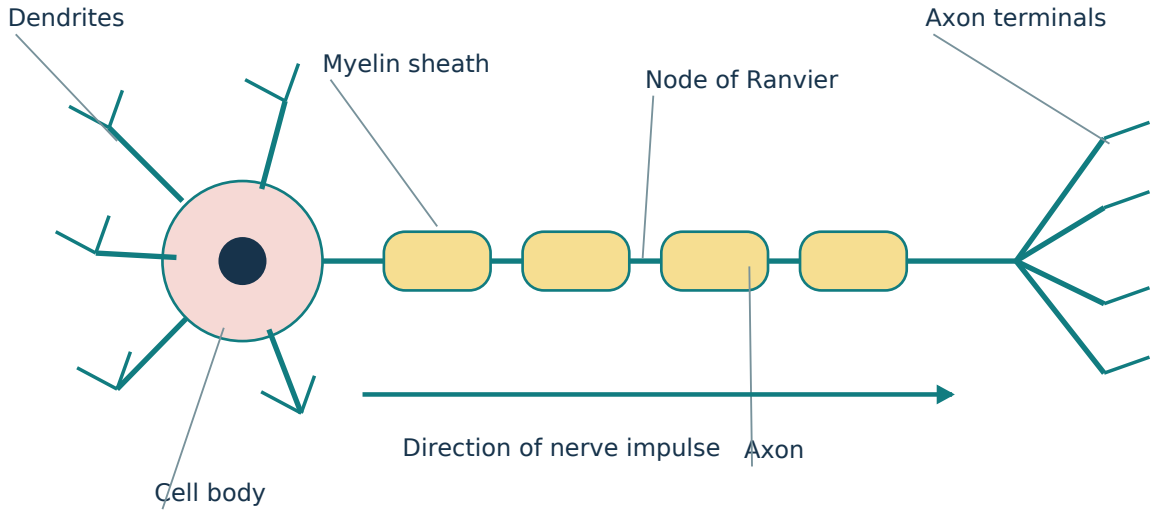
Q.2(v) OR | Neuron and the importance of Y and Z [3 marks]

A. The diagram shows a multipolar motor neuron, with many dendrites around the cell body and one long axon.

B. The yellow myelin sheath / Schwann-cell covering insulates the axon and reduces loss of electrical current. It allows much faster nerve transmission.

C. The nodes of Ranvier are gaps where the impulse is renewed, allowing it to travel rapidly from node to node. Damage to a node can slow or block transmission, causing weakness or loss of normal responses.

Reading the scan: The short Y/Z leaders lie near sheath-gap boundaries and are not perfectly sharp. Y is read here as the myelin covering and Z as a node. If a clearer original places a leader on the other structure, use the matching structure-function explanation above.



Q.2(vi) | Benefits of internal fertilisation in land animals [3 marks]

• Gametes meet inside the body, where they are protected from drying out.

• Sperm are placed near the eggs, increasing the chance of fertilisation and reducing gamete wastage.

• Fertilisation does not require an outside water medium, allowing successful reproduction on land.

Q.2(vi) OR | Components of gastric juice and their roles [3 marks]

• Hydrochloric acid: Makes the stomach acidic, helps kill swallowed microbes and activates pepsinogen into pepsin.

• Pepsin: Digests proteins into shorter peptides.

• Gastric lipase: Digests a small amount of fat. Mucus also protects the stomach wall from acid and enzymes; it does not digest food.

Q.2(vii) | Pea traits studied by Mendel [3 marks]

Trait	Dominant phenotype	Recessive phenotype
Plant height	Tall	Dwarf
Seed shape	Round	Wrinkled
Seed colour	Yellow	Green

Q.2(vii) OR | Completed pathogen table [3 marks]

Pathogen group	Example	Disease
Bacteria	Salmonella typhi	Typhoid
Fungi	Trichophyton species	Ringworm
Protozoa	Plasmodium species	Malaria

Q.2(viii) | Why some vaccines need booster shots [3 marks]

After vaccination, the amount of protective antibodies may fall with time, and protection may become weaker. A booster exposes memory lymphocytes to the antigen again. They multiply quickly and produce a stronger antibody response, improving and extending protection. Some vaccines also need repeated doses to build sufficient protection in the first place.

Q.2(viii) OR | Bt cotton and its importance in Pakistan [3 marks]

Bt cotton is genetically modified cotton containing gene(s) from the bacterium Bacillus thuringiensis. These genes allow the plant to make a protein toxic to certain insect pests, especially susceptible bollworms. In Pakistan, it can reduce insect damage and the need for some insecticide sprays, helping protect cotton yield and farmer income. Proper pest management is still needed because it does not kill every pest and resistance can develop.

Q.2(ix) | Biostatistics in public health [3 marks]

• It helps count disease cases and calculate rates, so outbreaks and high-risk groups can be identified.

• It helps compare the results of treatments and vaccination programmes to judge whether they work.

• It helps authorities plan hospitals, staff, medicines and budgets according to the health needs of the population.

Q.2(ix) OR | Flow sheet of sperm formation [3 marks]

Spermatogonium → primary spermatocyte → two secondary spermatocytes → four spermatids → four spermatozoa.

Stage	Process / chromosome number
Spermatogonium (2n)	Mitosis produces more spermatogonia; some grow.
Primary spermatocyte (2n)	Undergoes meiosis I.
Two secondary spermatocytes (n)	Undergo meiosis II.
Four spermatids (n)	Develop heads, middle pieces and tails (spermiogenesis).
Four sperm cells / spermatozoa (n)	Mature male gametes. In humans: 2n = 46; n = 23.

Q.2(x) | Adaptive immunity and defence against infection [3 marks]

Adaptive immunity is a specific defence that develops after exposure to an antigen, for example during infection or vaccination. B lymphocytes form plasma cells that make antibodies against the antigen. T lymphocytes coordinate responses or destroy infected cells. Memory cells remain and produce a faster, stronger response on later exposure to the same pathogen.

Q.2(x) OR | Alveoli are suited to gaseous exchange [3 marks]

• Millions of alveoli provide a very large surface area.

• Their moist walls and the adjacent capillary walls are only one cell thick, giving a short diffusion distance.

• A rich capillary supply and continuous ventilation maintain differences in oxygen and carbon dioxide concentration, allowing rapid diffusion.

Q.2(xi) | Why veins have valves but arteries do not [3 marks]

Blood in most veins flows at low pressure, so it may tend to move backwards, especially against gravity in the limbs. Venous valves stop this backflow and help muscle movements push blood towards the heart. Arteries carry blood away from the heart at much higher pressure and do not need valves along their length. The semilunar valves at the exits of the heart prevent return into the ventricles.

Q.2(xi) OR | In-situ and ex-situ bioremediation [3 marks]

Feature	In-situ	Ex-situ
Place of treatment	At the polluted site without removing the material.	After polluted material is removed to another treatment area.
Method	Supply microbes, oxygen or nutrients directly to the site.	Treat excavated soil or pumped water in tanks or prepared beds.
Example	Stimulating oil-degrading bacteria in contaminated soil in place.	Treating excavated oil-polluted soil in a bioreactor.

Paper 1: Model Paper – Section C

Q.3 to Q.6 carry 5 marks each. Both alternatives are answered. Use the labelled diagrams with the written explanation.

Q.3 | Types and roles of white blood cells [5 marks]

White blood cells have nuclei and defend the body. The five main types are:

Type and structure	Main function
Neutrophil: nucleus with several lobes; fine granules.	Engulfs and digests bacteria by phagocytosis.
Eosinophil: usually a two-lobed nucleus; large granules.	Helps destroy parasites and is involved in allergic responses.
Basophil: coarse granules often hide the nucleus.	Releases histamine during inflammation and heparin, an anticoagulant.
Monocyte: largest WBC; kidney-shaped nucleus.	Enters tissues and becomes a macrophage, engulfing pathogens and dead cells.
Lymphocyte: large round nucleus with little cytoplasm.	B cells form antibody-producing plasma cells. T cells kill infected cells or coordinate immunity; memory cells give long-term protection.

Q.3 OR | Nephron structure related to urine formation [5 marks]

A nephron is the microscopic structural and functional unit of a kidney. Each kidney contains about one million nephrons.

• Renal corpuscle: A cup-shaped, double-walled Bowman's capsule surrounds a network of blood capillaries called the glomerulus. The afferent arteriole brings blood in; the efferent arteriole carries it out.

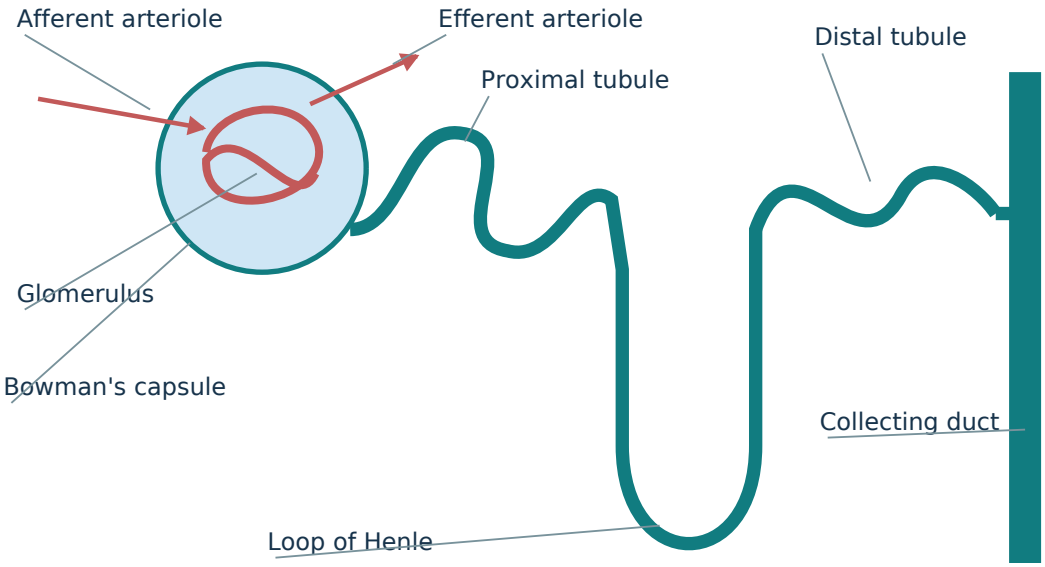
• Renal tubule: The capsule opens into the proximal convoluted tubule, followed by the descending and ascending limbs of the loop of Henle and the distal convoluted tubule.

• Collecting duct: Receives fluid from several nephrons and carries urine towards the renal pelvis. A capillary network surrounds the tubule. The corpuscle and convoluted tubules lie mainly in the cortex; loops extend into the medulla.

• 1. Glomerular filtration: Blood pressure forces water, urea, salts, glucose and other small solutes into Bowman's capsule. Blood cells and most proteins remain in the blood.

• 2. Selective reabsorption: Useful glucose, amino acids, much water and needed salts return from the tubule to the blood. The proximal tubule does most of this work. The loop helps build a concentration gradient.

• 3. Tubular secretion and concentration: Extra hydrogen ions, potassium ions and some drugs pass from blood into the tubule. Regulated water reabsorption in the distal tubule and collecting duct helps control the final urine concentration.



Q.4 | Structure of a villus and absorption [5 marks]

Structure: A villus is a tiny finger-like projection in the lining of the small intestine. It has an epithelium only one cell thick, many blood capillaries and a central lymph vessel called a lacteal. Its cells have microvilli.

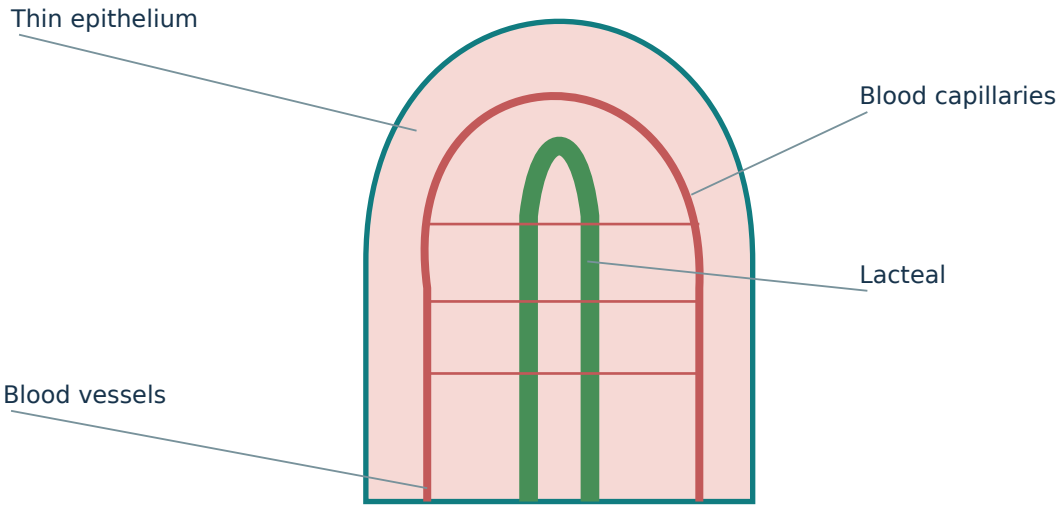
• Large surface: Many villi and microvilli provide a large area for absorption.

• Short distance: The thin wall allows nutrients to pass across quickly.

• Blood capillaries: Absorb glucose, amino acids, water-soluble vitamins and mineral ions; these travel towards the liver in the hepatic portal vein.

• Lacteal: Absorbs most digested fats after they are rebuilt and packed into tiny particles in intestinal cells. Fat enters lymph and later the blood.

• Continuous flow: Blood and lymph carry nutrients away and maintain a concentration gradient. Nutrients cross by diffusion or active transport, while water follows by osmosis.



Finger-like villus; microvilli add more surface area

Q.4 OR | Law of independent assortment and dihybrid cross [5 marks]

Law of independent assortment: During gamete formation, alleles of different genes assort independently when the genes are unlinked. Thus inheritance of one trait does not determine inheritance of the other.

Symbols: R = round seed; r = wrinkled seed; Y = yellow seed; y = green seed. R and Y are dominant.

Parental cross: RRYY (round yellow) × rryy (wrinkled green). Gametes: RY and ry. All F_1 offspring are RrYy, round and yellow.

F_1 self-cross: RrYy × RrYy. Each parent forms four gamete types in equal proportions: RY, Ry, rY and ry.

Gametes	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

F_2 phenotypic ratio: 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green. The new combinations, round green and wrinkled yellow, show independent assortment. This expected ratio assumes independent genes and complete dominance.

Q.5 | Biotechnology in food crops and dairy livestock in Pakistan [5 marks]

A. Improvement of food crops

• Tissue culture: Produces many identical, healthy plants quickly; it can provide disease-free planting material such as potato stocks.

• Crop improvement: Genetic engineering and marker-assisted breeding help develop crops with useful traits such as pest resistance, stress tolerance and improved nutritional quality.

• Biological inputs: Biofertilisers improve nutrient availability, while biopesticides help control pests. These can support yield and reduce some chemical inputs.

B. Improvement of dairy livestock

• Artificial insemination: Semen from selected high-quality bulls can spread useful inherited traits, such as higher milk yield, through cattle and buffalo populations.

• Embryo transfer: Embryos from valuable females can be placed in other females to increase the number of offspring from superior breeding stock.

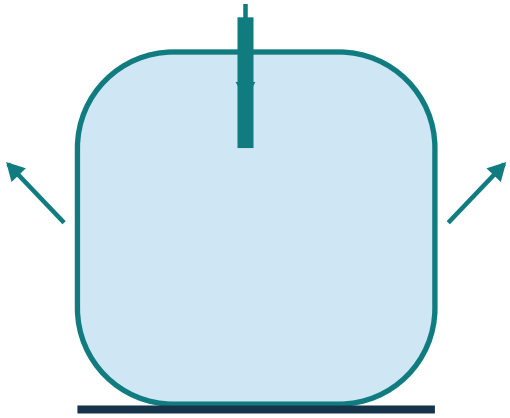
• Animal health and selection: Diagnostic tests, vaccines and genetic selection help improve herd health, productivity and milk quality.

In Pakistan these methods can improve food supply and farmer income when used with suitable breeds and good management. Bt cotton is an important agricultural example, but cotton is not a food-crop example.

Q.5 OR | Changes during inhalation and exhalation [5 marks]

Change	Inhalation (breathing in)	Exhalation (quiet breathing out)
Intercostal muscles	External intercostal muscles contract.	External intercostal muscles relax.
Ribs	Move upward and outward.	Move downward and inward.
Diaphragm	Contracts and becomes flatter.	Relaxes and becomes dome-shaped.
Chest / lung volume	Increases; lungs expand.	Decreases; elastic lungs recoil.
Pressure and air	Pressure falls below atmospheric pressure; air enters.	Pressure rises above atmospheric pressure; air leaves.

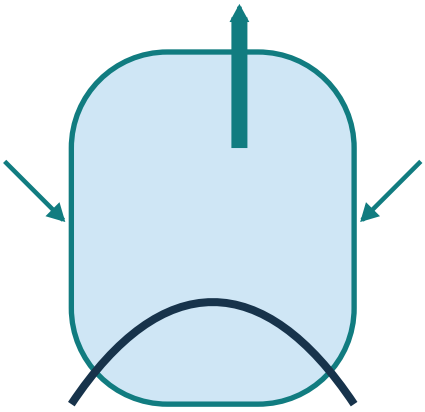
INHALATION



Diaphragm flattens

Ribs up and outward

EXHALATION



Diaphragm domes up

Ribs down and inward

Q.6 | Transmission of a nerve impulse through a neuron [5 marks]

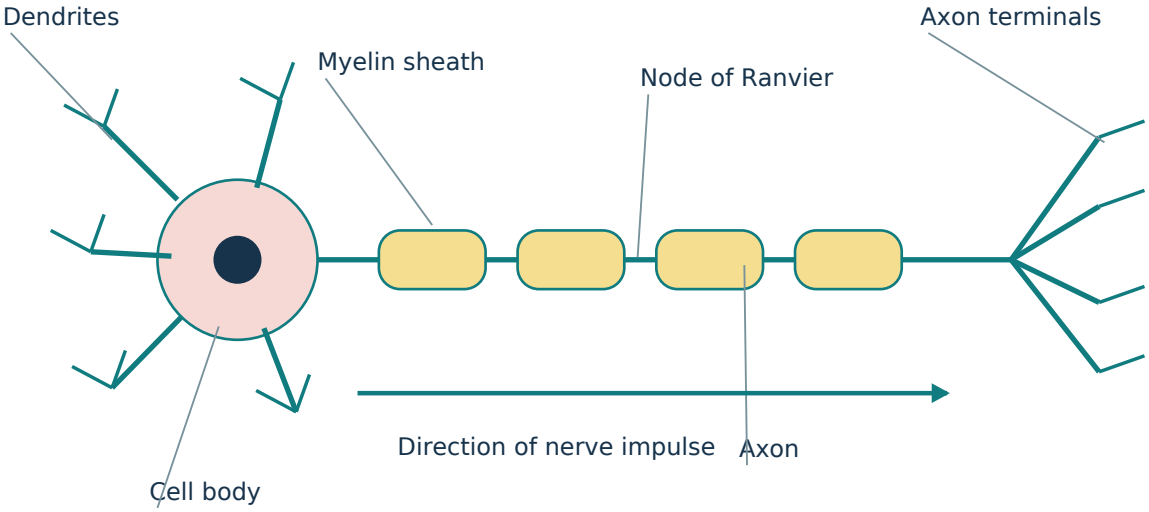
• Resting state: The axon membrane is polarised: its inside is negative compared with the outside. Ion gradients are maintained partly by the sodium-potassium pump.

• Depolarisation: A stimulus strong enough to reach threshold opens sodium channels. Sodium ions enter, briefly making the inside more positive. This forms an action potential.

• Propagation: Local electrical changes stimulate the next part of the axon membrane, so the impulse travels along the axon. In a myelinated axon, it is renewed at nodes of Ranvier and travels rapidly between them.

• Repolarisation: Sodium channels close and potassium channels open. Potassium leaves, returning the inside towards its negative state. A short refractory period helps ensure forward transmission.

• At the terminal: The impulse causes neurotransmitter release into a synapse; this may stimulate the next neuron or an effector.



Q.6 OR | Mean and mode of game scores [5 marks]

Mean: The sum of all observations divided by their number.
Mode: The observation that occurs most often.

Data: 5, 10, 4, 8, 11, 4, 7.
Sum = 5 + 10 + 4 + 8 + 11 + 4 + 7 = 49.
Number of students = 7.
Mean = 49 ÷ 7 = 7.
The score 4 occurs twice; each other score occurs once.
Mode = 4.

Paper 2: 1st Annual 2026 - Variant B

Source: uploaded PDF page 3; printed paper code 2SA-I 26006 (B).

Section B: Q.2 (33 marks) - Both choices of every part are answered below for practice. In the examination, follow the choice instructions printed on your paper.

Q.2(i) | Three types of neurons [3 marks]

• Sensory neuron: Carries impulses from receptors, such as those in the skin, towards the central nervous system.

• Relay neuron (interneuron): Lies within the central nervous system and passes impulses between other neurons, such as sensory and motor neurons.

• Motor neuron: Carries impulses from the central nervous system to an effector, such as a muscle or gland.

Q.2(i) OR | Stimulus and two examples [3 marks]

A stimulus is a detectable change in the internal or external environment that can cause a response. Example 1: Heat from a hot object stimulates skin receptors, causing withdrawal of the hand. Example 2: Bright light stimulates receptors in the eye, causing the pupil to become smaller.

Q.2(ii) | Reproduction shown in Hydra [3 marks]

The method is budding, a form of asexual reproduction. Repeated mitotic divisions form a small outgrowth, or bud, on the parent Hydra. The bud grows and develops a mouth and tentacles. When mature, it separates and lives independently. It is genetically very similar to the parent unless a mutation occurs.

Q.2(ii) OR | Flower labels and one function each [3 marks]

Label	Part	Function
A	Anther	Produces pollen grains containing the male gametes.
B	Petal	Helps attract pollinators by colour, scent or shape.
C	Ovary	Contains and protects the ovules; after fertilisation the ovary develops into a fruit.

Q.2(iii) | Cross of TT and tt pea plants [3 marks]

Parents: TT (tall) × tt (dwarf).
Gametes: The TT parent produces T gametes; the tt parent produces t gametes.
F_1 genotype: All offspring are Tt (100% heterozygous).
F_1 phenotype: All offspring are tall, because T is dominant over t.

Gametes	T	T
t	Tt	Tt
t	Tt	Tt

Q.2(iii) OR | Food biotechnology and a GM crop in Pakistan [3 marks]

Food biotechnology is the use of living organisms, cells, enzymes or genetic methods to improve food production, processing, quality or preservation. It includes fermentation, tissue culture and genetic improvement. One genetically modified crop grown in Pakistan is Bt cotton, which resists certain insect pests. It is a GM crop example, although cotton is not normally classified as a food crop.

Q.2(iv) | Three roles of the immune system [3 marks]

• Recognises invaders: Identifies foreign antigens and harmful microorganisms.

• Destroys pathogens: Phagocytes engulf germs, antibodies neutralise them, and cytotoxic T cells destroy infected cells.

• Provides memory: Memory lymphocytes produce a faster, stronger response if the same pathogen returns.

Q.2(iv) OR | Role of liver in digestion [3 marks]

• The liver produces bile, which is stored and concentrated in the gall bladder and released into the small intestine.

• Bile salts emulsify fat: They break large fat globules into smaller droplets, increasing the surface area for lipase.

• Bile is alkaline and helps neutralise acidic chyme from the stomach, providing suitable conditions for intestinal enzymes. Bile has no digestive enzymes.

Q.2(v) | Treatment and prevention of myocardial infarction [3 marks]

Treatment: It requires urgent hospital treatment to restore blood flow to the heart muscle. Depending on the case, doctors use clot-dissolving drugs or coronary angioplasty, often with a stent; bypass surgery is used in selected cases.

• Prevention 1: Avoid smoking and tobacco, which damage blood vessels.

• Prevention 2: Take regular exercise and eat a balanced diet low in excess saturated fat and salt to help control weight, blood pressure and cholesterol.

Q.2(v) OR | Gene and allele with examples [3 marks]

Term	Meaning	Example
Gene	A section of DNA that carries information for a functional product and can influence a characteristic.	A gene influencing pea-plant height.
Allele	An alternative form of the same gene, at the same position on homologous chromosomes.	T (tall) and t (dwarf) are alleles of the height gene.

Q.2(vi) | Bronchitis: definition, cause and symptom [3 marks]

Definition: Bronchitis is inflammation of the lining of the bronchi, often with excess mucus.
One cause: Cigarette smoke irritates the bronchi and can cause chronic bronchitis. Acute bronchitis is commonly caused by a viral infection.
One symptom: A persistent cough, often bringing up mucus.

Q.2(vi) OR | Sympathetic and parasympathetic systems [3 marks]

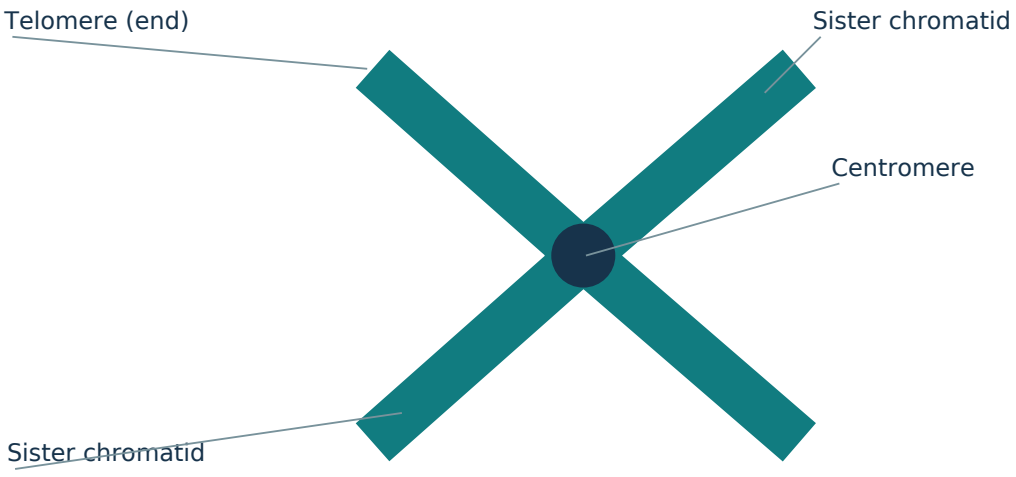
Feature	Sympathetic	Parasympathetic
Main role	Prepares the body for emergency: fight or flight.	Supports normal rest and digestion.
Heart and breathing	Increases heart rate; dilates airways.	Slows heart rate; tends to narrow airways towards resting size.
Digestion and pupil	Reduces digestive activity; dilates pupils.	Increases digestive activity; constricts pupils.

Q.2(vii) | Spermatogenesis [3 marks]

Spermatogenesis is the formation of sperm in the seminiferous tubules of the testes. Diploid spermatogonia divide by mitosis; some grow into primary spermatocytes. Each primary spermatocyte undergoes meiosis I and II to form four haploid spermatids. These change into sperm with a head, middle piece and tail.

Q.2(vii) OR | Labelled chromosome [3 marks]

A duplicated chromosome has two sister chromatids joined at the centromere. Each chromatid contains a DNA molecule associated with proteins. The ends are called telomeres.



Duplicated chromosome: two sister chromatids

Q.2(viii) | Medical benefits of marine biotechnology [3 marks]

• Anticancer medicines: Marine organisms supply compounds from which some anticancer drugs have been developed.

• Antimicrobial compounds: Marine bacteria, fungi and other organisms provide substances that act against certain disease-causing microbes.

• Useful health products: Marine sources provide omega-3 oils and materials such as alginates used in wound dressings and drug delivery. These products support healthcare.

Q.2(viii) OR | Blood clotting [3 marks]

At an injury, platelets stick to the damaged vessel and form a temporary plug. Platelets and damaged tissues start a clotting cascade. This forms thrombin from prothrombin, with calcium ions and other clotting factors involved. Thrombin changes soluble fibrinogen into insoluble fibrin. Fibrin threads trap blood cells and form a clot that seals the wound.

Q.2(ix) | Three pathogen groups and one disease each [3 marks]

Pathogen group	Example disease	Causative agent
Bacteria	Typhoid	Salmonella typhi
Viruses	COVID-19	SARS-CoV-2
Fungi	Ringworm	Trichophyton species

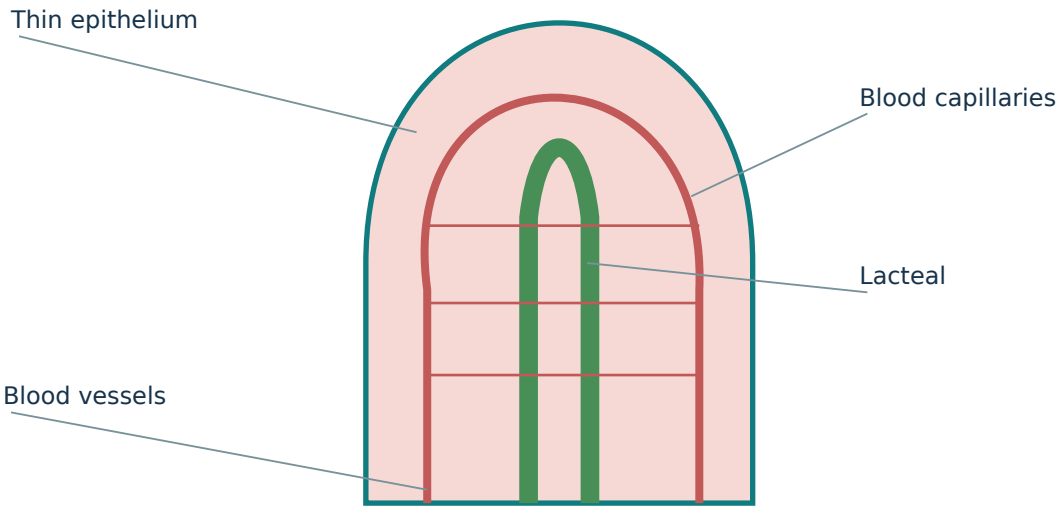
Q.2(ix) OR | Mean and mode of monthly patients [3 marks]

Data (January to June): 9, 13, 22, 14, 22, 20.
Sum = 9 + 13 + 22 + 14 + 22 + 20 = 100.
Mean = 100 ÷ 6 = 16.67 patients per month (approximately).
Mode = 22 patients, because 22 occurs twice, more often than any other value.

Q.2(x) | Structure of a villus [3 marks]

Structure: A villus is a tiny finger-like projection in the lining of the small intestine. It has an epithelium only one cell thick, many blood capillaries and a central lymph vessel called a lacteal. Its cells have microvilli.

The large surface and thin wall allow rapid absorption. Blood capillaries carry sugars and amino acids away; the lacteal carries absorbed fats.



Finger-like villus; microvilli add more surface area

Q.2(x) OR | Composition of inspired and expired air [3 marks]

Component	Inspired air	Expired air
Oxygen	About 21%	About 16%
Carbon dioxide	About 0.04%	About 4%
Nitrogen	About 78%	About 78% (little change)
Water vapour	Variable; usually less	More; nearly saturated
Temperature	Depends on surroundings	Usually warmer, near body temperature

Q.2(xi) | Roles of red blood cells [3 marks]

• Oxygen transport: Haemoglobin binds oxygen in the lungs and releases it in body tissues.

• Carbon dioxide transport: Red blood cells help convert much of the carbon dioxide into bicarbonate; some CO_2 also binds to haemoglobin.

• Acid-base balance: Haemoglobin helps buffer hydrogen ions, reducing large changes in blood pH.

Q.2(xi) OR | Glomerular filtration [3 marks]

Glomerular filtration is the pressure-driven movement of water and small dissolved substances from glomerular blood into Bowman's capsule. Blood enters through the afferent arteriole; the glomerular arrangement maintains high filtration pressure. Water, glucose, amino acids, salts and urea pass through the filtration barrier. Blood cells and most plasma proteins are too large to pass, so they remain in the blood.

Paper 2: 1st Annual 2026 - Variant B - Section C

Q.3 to Q.6 carry 5 marks each. Both alternatives are answered. Use the labelled diagrams with the written explanation.

Q.3 | Master gland: parts and functions [5 marks]

The pituitary gland, located below the hypothalamus at the base of the brain, is called the master gland because several of its hormones control other endocrine glands. The hypothalamus controls the pituitary.

1. Anterior pituitary: This glandular part produces and secretes several hormones.

Hormone	Main function
Growth hormone (GH)	Promotes growth of bones and body tissues.
Thyroid-stimulating hormone (TSH)	Stimulates the thyroid to release thyroid hormones.
Adrenocorticotropic hormone (ACTH)	Stimulates the adrenal cortex.
Follicle-stimulating hormone (FSH)	Supports egg-follicle development in ovaries and sperm production in testes.
Luteinising hormone (LH)	Triggers ovulation and supports ovarian hormones; stimulates testosterone secretion in males.
Prolactin	Stimulates milk production after childbirth.

2. Posterior pituitary: This neural part stores and releases hormones made in the hypothalamus. ADH increases water reabsorption by the kidneys. Oxytocin causes uterine contractions during childbirth and milk ejection during feeding.

Q.3 OR | Diabetes, its types and effects [5 marks]

Diabetes mellitus is a disease in which blood glucose stays higher than normal because insulin is absent, insufficient or not used properly.

Feature	Type 1 diabetes	Type 2 diabetes
Main problem	Immune destruction of pancreatic beta cells; little or no insulin.	Body cells respond poorly to insulin; insulin supply may also become insufficient.
Usual pattern	Often begins in childhood or youth, but can occur at any age.	More common in adults; can also develop in young people.
Control	Insulin treatment, balanced diet, exercise and glucose monitoring.	Healthy diet, activity, weight management and medicines; insulin may also be needed.

Effects: High glucose causes frequent urination, excessive thirst, tiredness and sometimes weight loss. Over time it can damage blood vessels, nerves, kidneys and the retina, and delay wound healing.

Q.4 | Biostatistics and four applications [5 marks]

Biostatistics is the use of statistical methods to collect, organise, analyse and interpret data in biology, medicine and public health.

• Disease monitoring: Count cases and compare disease rates in different places or time periods.

• Testing treatments: Compare results in groups to assess the effectiveness and safety of medicines or vaccines.

• Health planning: Use patient data to plan beds, staff, medicines and health budgets.

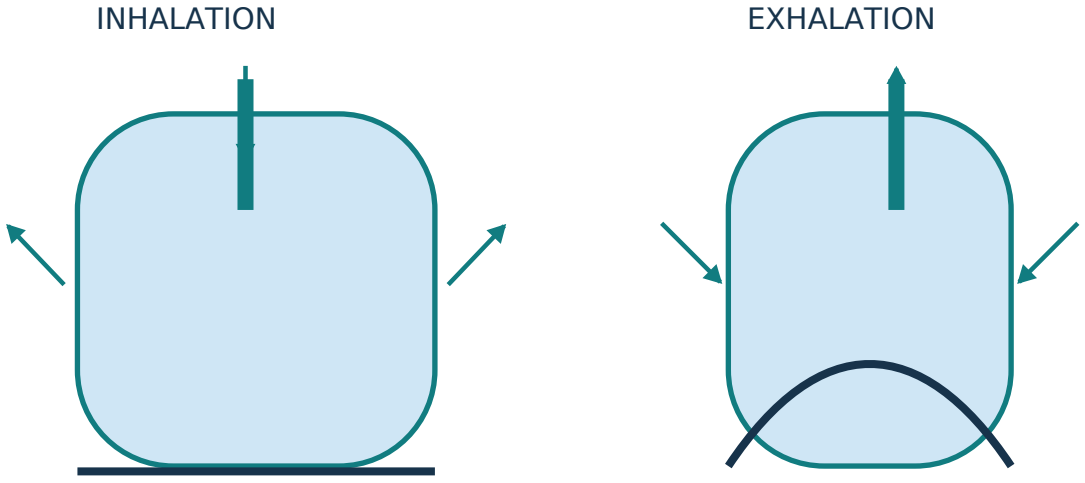
• Risk factors: Study links between factors such as smoking and disease.

• Agriculture and research: Compare crop yields, growth or experimental results to reach evidence-based conclusions.

Any four of the five applications above, explained clearly, answer the application part.

Q.4 OR | Mechanism of breathing with diagrams [5 marks]

Change	Inhalation (breathing in)	Exhalation (quiet breathing out)
Intercostal muscles	External intercostal muscles contract.	External intercostal muscles relax.
Ribs	Move upward and outward.	Move downward and inward.
Diaphragm	Contracts and becomes flatter.	Relaxes and becomes dome-shaped.
Chest / lung volume	Increases; lungs expand.	Decreases; elastic lungs recoil.
Pressure and air	Pressure falls below atmospheric pressure; air enters.	Pressure rises above atmospheric pressure; air leaves.



Diaphragm flattens
Ribs up and outward

Diaphragm domes up
Ribs down and inward

Q.5 | Industrial biotechnology: concept and applications [5 marks]

Concept: Industrial biotechnology uses microorganisms, cells or enzymes in large-scale processes to make useful products. Selected organisms grow in controlled fermenters with suitable temperature, pH, nutrients and oxygen conditions.

• Food processing: Yeast ferments sugars to make carbon dioxide for bread and alcohol for fermentation products. Lactic acid bacteria are used in yoghurt.

• Medicines and enzymes: Microbial fermentation produces antibiotics, and microbes make enzymes used in detergents and food processing.

• Industrial chemicals and fuels: Microorganisms can produce organic acids, such as citric acid, and fuels such as bioethanol or biogas.

Basic process: Select organism → prepare nutrient medium → grow it in a controlled fermenter → collect and purify the required product. Biotechnology can use renewable raw materials and reduce waste in suitable processes.

Q.5 OR | Human stomach: structure and functions [5 marks]

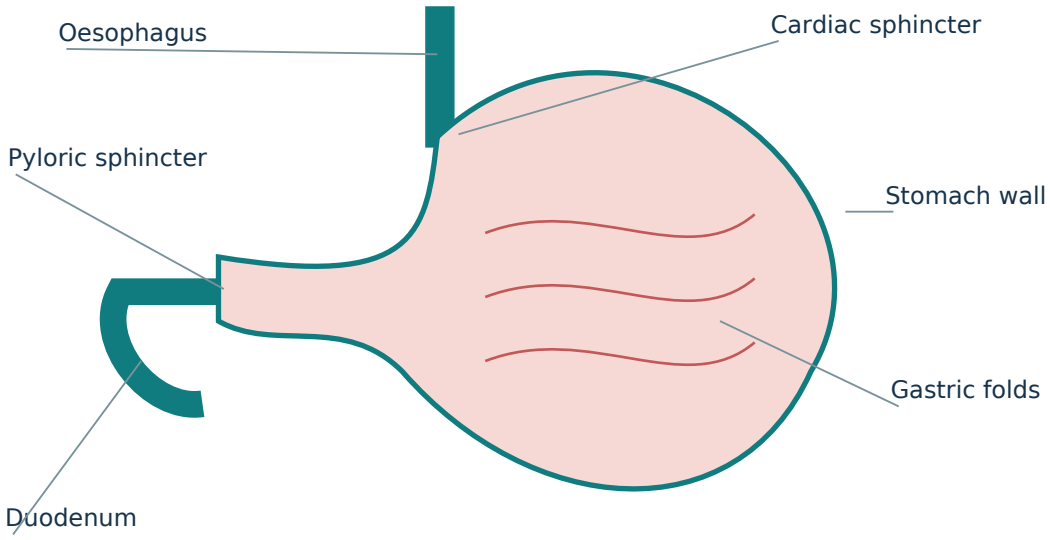
Structure: The stomach is a J-shaped muscular sac between the oesophagus and duodenum. A cardiac/lower oesophageal sphincter controls entry and the pyloric sphincter controls exit. Its muscular wall has longitudinal, circular and oblique layers. The folded inner lining, called rugae, contains gastric glands.

• Storage and mixing: The stomach temporarily stores food. Muscular contractions churn it and mix it with gastric juice to form semi-fluid chyme.

• Protein digestion: Gastric glands release pepsinogen, which becomes pepsin in acid. Pepsin breaks proteins into shorter peptides.

• Acid and protection: Hydrochloric acid provides a suitable pH and helps kill swallowed microbes. Mucus protects the stomach lining.

• Controlled release: The pyloric sphincter releases small amounts of chyme into the duodenum for further digestion.



Q.6 | Structure and functions of WBC types [5 marks]

White blood cells have nuclei and defend the body. The five main types are:

Type and structure	Main function
Neutrophil: nucleus with several lobes; fine granules.	Engulfs and digests bacteria by phagocytosis.
Eosinophil: usually a two-lobed nucleus; large granules.	Helps destroy parasites and is involved in allergic responses.
Basophil: coarse granules often hide the nucleus.	Releases histamine during inflammation and heparin, an anticoagulant.
Monocyte: largest WBC; kidney-shaped nucleus.	Enters tissues and becomes a macrophage, engulfing pathogens and dead cells.
Lymphocyte: large round nucleus with little cytoplasm.	B cells form antibody-producing plasma cells. T cells kill infected cells or coordinate immunity; memory cells give long-term protection.

Q.6 OR | Dialysis, haemodialysis and peritoneal dialysis [5 marks]

Dialysis: The removal of waste substances and excess water and salts from the blood when kidneys cannot do this adequately. A selectively permeable membrane separates blood from dialysis fluid.

Haemodialysis: Blood is taken through tubing to a dialyser outside the body. Small wastes, such as urea, diffuse across its membrane into dialysis fluid. Extra water is removed by controlled pressure. Blood cells and most proteins stay in the blood, which is returned to the patient.

Peritoneal dialysis: Dialysis fluid is placed in the abdominal cavity through a catheter. The lining of the abdomen, the peritoneum, acts as the membrane. Wastes and extra water pass from blood capillaries into this fluid. After a set time, the used fluid is drained and replaced.

Common principle: Both methods remove small wastes across a membrane. Haemodialysis uses an artificial dialyser; peritoneal dialysis uses the body's peritoneum.

Paper 3: 1st Annual 2026 - Variant D

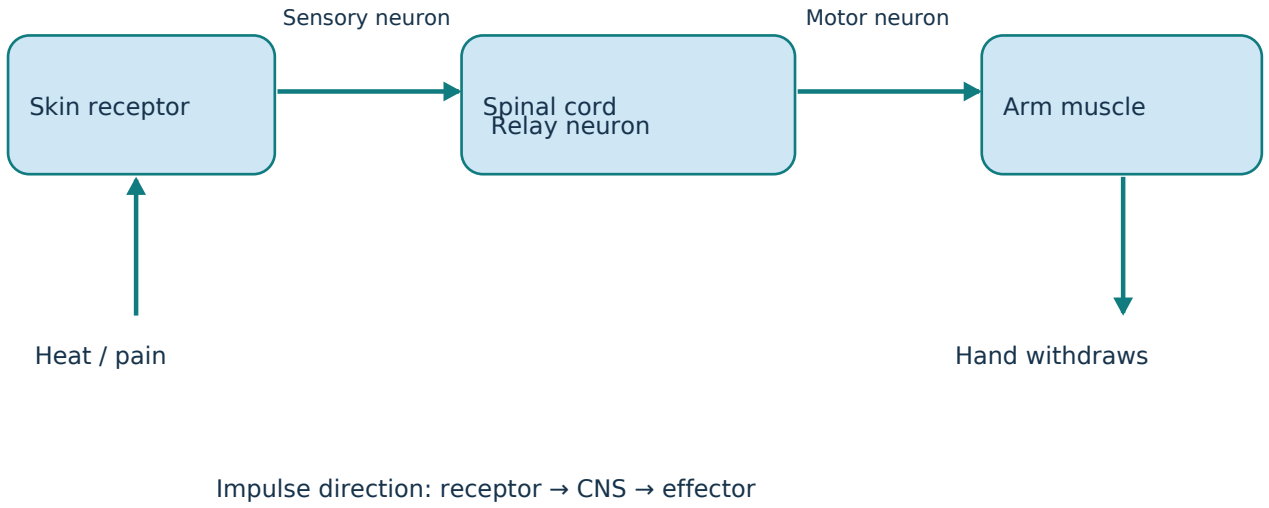
Source: uploaded PDF page 4; printed paper code 2SA-I 26006 (D).

Section B: Q.2 (33 marks) - Both choices of every part are answered below for practice. In the examination, follow the choice instructions printed on your paper.

Q.2(i) | Withdrawal reflex: pathway of the nerve impulse [3 marks]

Heat / pain stimulus → receptor in the skin → sensory neuron → relay neuron in spinal cord → motor neuron → arm flexor muscle (effector) → muscle contracts and hand withdraws.

Step	Sketch pathway / action
1. Skin receptor	Detects heat or pain from the hot object.
2. Sensory neuron	Carries the impulse into the spinal cord.
3. Relay neuron	Transfers the impulse across synapses to a motor neuron.
4. Motor neuron	Carries the impulse out to the arm muscle.
5. Effector muscle	Contracts and pulls the hand away.



Impulse direction: receptor → CNS → effector

The spinal cord coordinates this rapid protective reflex. The brain also receives information, so the person becomes aware of the pain.

Q.2(i) OR | Two plant stimuli and responses [3 marks]

• Light: A shoot bends towards light. This is positive phototropism and helps the leaves receive light for photosynthesis.

• Gravity: Roots grow downward in the direction of gravity. This is positive gravitropism and helps the plant obtain water and minerals and anchor itself.

Q.2(ii) | Amoeba reproduction in the diagram [3 marks]

The diagram shows binary fission, an asexual method of reproduction in Amoeba. First the genetic material is copied. The nucleus divides into two by mitosis (karyokinesis). The cytoplasm then divides (cytokinesis), producing two separate daughter amoebae. Each receives a nucleus and can grow independently.

Q.2(ii) OR | Cross of two heterozygous tall plants [3 marks]

Parents: Tt × Tt. Each parent produces T and t gametes.

Gametes	T	t
T	TT	Tt
t	Tt	tt

Genotypic ratio: 1 TT : 2 Tt : 1 tt.
Phenotypic ratio: 3 tall : 1 dwarf. Both TT and Tt are tall because T is dominant.

Q.2(iii) | Homozygous and heterozygous conditions [3 marks]

Condition	Meaning	Example
Homozygous	Both alleles of a gene are the same.	TT (pure tall) or tt (pure dwarf).
Heterozygous	The two alleles of a gene are different.	Tt: tall because T is dominant over t.

Q.2(iii) OR | Importance of Penicillium notatum [3 marks]

• Discovery: In 1928, Alexander Fleming noticed that the mould Penicillium notatum prevented bacteria from growing around it.

• Antibiotic: The mould produced penicillin, which stops susceptible bacteria from making proper cell walls.

• Importance: Penicillin made many previously dangerous bacterial infections treatable, saved lives and encouraged the development of other antibiotics. It does not treat viral infections.

Q.2(iv) | Two examples of agricultural biotechnology in Pakistan [3 marks]

• Bt cotton: Genetically modified plants produce a Bt protein that protects against certain bollworms, reducing crop losses.

• Plant tissue culture: Healthy disease-free planting material, such as potato stocks and banana plants, can be multiplied quickly. This improves planting quality and supports crop production.

Q.2(iv) OR | Liver functions other than digestion [3 marks]

• Blood glucose regulation: Converts extra glucose into glycogen and releases glucose when needed.

• Detoxification: Processes many drugs, alcohol and harmful substances.

• Protein metabolism: Converts toxic ammonia from amino-acid breakdown into urea. It also makes important plasma proteins, including albumin and fibrinogen.

Q.2(v) | Blood transport of oxygen and carbon dioxide [3 marks]

Oxygen: In the lungs, oxygen diffuses into the blood and binds reversibly to haemoglobin in RBCs, forming oxyhaemoglobin. It is released in tissues where oxygen concentration is lower.

Carbon dioxide: It diffuses from tissues into blood. Most is carried as bicarbonate ions after conversion mainly inside RBCs; some binds to haemoglobin and a small amount dissolves in plasma. In the lungs these changes reverse and CO_2 is exhaled.

Q.2(v) OR | Three roles of white blood cells [3 marks]

• Phagocytosis: Neutrophils and macrophages engulf and digest pathogens and cell debris.

• Specific defence: B lymphocytes make antibody-producing cells; T lymphocytes help coordinate defence or kill infected cells.

• Memory and inflammation: Memory lymphocytes give quicker future protection; other WBCs release substances that help organise inflammation.

Q.2(vi) | Pneumonia: meaning, cause and symptom [3 marks]

Meaning: Pneumonia is an infection causing inflammation of lung alveoli, which may fill with fluid or pus and reduce gaseous exchange.
One cause: Infection with the bacterium Streptococcus pneumoniae.
One symptom: Cough with fever; breathing difficulty may also occur.

Q.2(vi) OR | Cerebellum and medulla oblongata [3 marks]

Cerebellum: Coordinates voluntary muscle movements and maintains balance and posture. It makes movements smooth and accurate.

Medulla oblongata: Controls important involuntary activities such as breathing, heart rate and blood-vessel diameter. It also controls reflexes such as swallowing, coughing and vomiting.

Q.2(vii) | Median and mean of fruit production [3 marks]

Data: 3, 5, 18, 21, 15, 10, 8, 12, 13, 7.
Ordered data: 3, 5, 7, 8, 10, 12, 13, 15, 18, 21.
There are 10 values, so the median is the average of the 5th and 6th values.
Median = (10 + 12) ÷ 2 = 11 fruits.
Sum = 3 + 5 + 18 + 21 + 15 + 10 + 8 + 12 + 13 + 7 = 112.
Mean = 112 ÷ 10 = 11.2 fruits per tree.

Q.2(vii) OR | Blood-vessel diagram labels [3 marks]

Label	Part	One function
A	Artery / arteriole (from heart)	Carries blood away from the heart towards tissue capillaries.
B	Vein / venule (to heart)	Returns blood from tissue capillaries towards the heart.
C	Capillary network	Allows exchange of gases, nutrients and wastes between blood and tissue cells.

Q.2(viii) | One disease caused by a virus [3 marks]

COVID-19 is caused by the virus SARS-CoV-2. It spreads mainly through infectious respiratory particles in the air. Common symptoms include fever, cough, sore throat and tiredness. Severe disease can cause pneumonia and reduced blood oxygen. Vaccination, cleaner indoor air and appropriate hygiene help reduce its effects and spread.

Q.2(viii) OR | Role of platelets in clotting [3 marks]

Platelets stick to the damaged vessel wall, become activated and clump together to form a temporary platelet plug. They provide a surface and release substances that support the clotting cascade. This leads to formation of thrombin, which changes fibrinogen into fibrin. The fibrin mesh traps blood cells and strengthens the plug into a clot.

Q.2(ix) | Marine biotechnology in food production [3 marks]

• Aquaculture: Biotechnology helps breed and rear fish, shrimps and other seafood under controlled conditions, increasing supply.

• Health and growth: Improved feeds, diagnostic tests and disease control reduce losses and improve production.

• Algal products: Seaweeds and microalgae supply edible products, proteins and oils; algal substances such as agar and alginates are also used in food processing.

Q.2(ix) OR | Spermatogenesis [3 marks]

Spermatogenesis is the formation of sperm in the seminiferous tubules of the testes. Diploid spermatogonia divide by mitosis; some grow into primary spermatocytes. Each primary spermatocyte undergoes meiosis I and II to form four haploid spermatids. These change into sperm with a head, middle piece and tail.

Q.2(x) | Lacteal and its function [3 marks]

A lacteal is a small lymph capillary present in the centre of an intestinal villus. It absorbs most dietary fats after they are rebuilt into triglycerides and packed into particles called chylomicrons inside intestinal cells. It carries these fats, and associated fat-soluble vitamins, through lymph towards the bloodstream.

Q.2(x) OR | Plasma and what it transports [3 marks]

Plasma is the pale-yellow liquid part of blood. It is mainly water containing dissolved proteins and other substances. It transports digested nutrients, hormones, carbon dioxide mainly as bicarbonate, urea and other wastes, mineral ions and heat. It also carries blood cells, antibodies and clotting proteins.

Q.2(xi) | Inspired versus expired air: O2, CO2 and water [3 marks]

Component	Inspired air	Expired air
Oxygen	About 21%	About 16%
Carbon dioxide	About 0.04%	About 4%
Water vapour	Variable; generally less	More; nearly saturated

Q.2(xi) OR | Selective reabsorption in a nephron [3 marks]

Selective reabsorption returns useful substances from the filtrate in the renal tubule to the blood. In the proximal tubule, all normally filtered glucose and amino acids and much of the salts are reabsorbed, using active transport and other transport processes. Water follows by osmosis. Further regulated reabsorption occurs along the loop, distal tubule and collecting duct, helping maintain normal water and salt balance.

Paper 3: 1st Annual 2026 - Variant D - Section C

Q.3 to Q.6 carry 5 marks each. Both alternatives are answered. Use the labelled diagrams with the written explanation.

Q.3 | Pancreas: endocrine and exocrine roles [5 marks]

The pancreas is a mixed gland. Its exocrine part releases pancreatic juice through a duct into the duodenum. Its endocrine part consists of clusters of cells called the islets of Langerhans, which release hormones directly into the blood.

Insulin: Beta cells release insulin when blood glucose rises, such as after a meal. Insulin increases glucose uptake in many cells and encourages the liver and muscles to store glucose as glycogen. It also supports fat storage and reduces glucose production by the liver. Therefore, it lowers blood glucose.

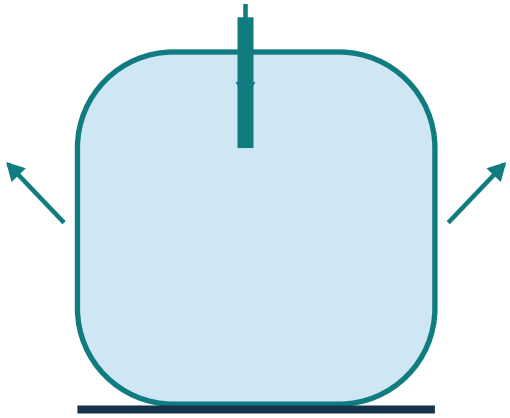
Glucagon: Alpha cells release glucagon when blood glucose falls. It stimulates the liver to break down glycogen into glucose and to form glucose from non-carbohydrate materials. Therefore, it raises blood glucose.

Negative feedback: Once glucose returns towards normal, secretion of the correcting hormone decreases. Insulin and glucagon work in opposite directions to keep blood glucose within a suitable range. Too little effective insulin causes diabetes mellitus.

Q.3 OR | Inhalation and exhalation [5 marks]

Change	Inhalation (breathing in)	Exhalation (quiet breathing out)
Intercostal muscles	External intercostal muscles contract.	External intercostal muscles relax.
Ribs	Move upward and outward.	Move downward and inward.
Diaphragm	Contracts and becomes flatter.	Relaxes and becomes dome-shaped.
Chest / lung volume	Increases; lungs expand.	Decreases; elastic lungs recoil.
Pressure and air	Pressure falls below atmospheric pressure; air enters.	Pressure rises above atmospheric pressure; air leaves.

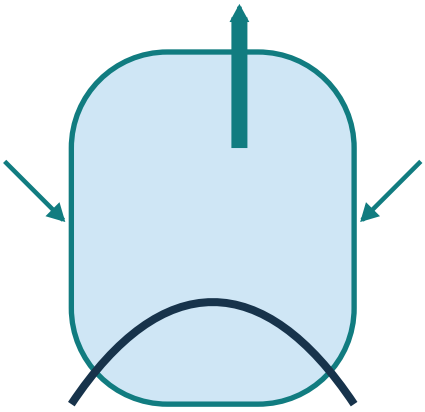
INHALATION



Diaphragm flattens

Ribs up and outward

EXHALATION



Diaphragm domes up

Ribs down and inward

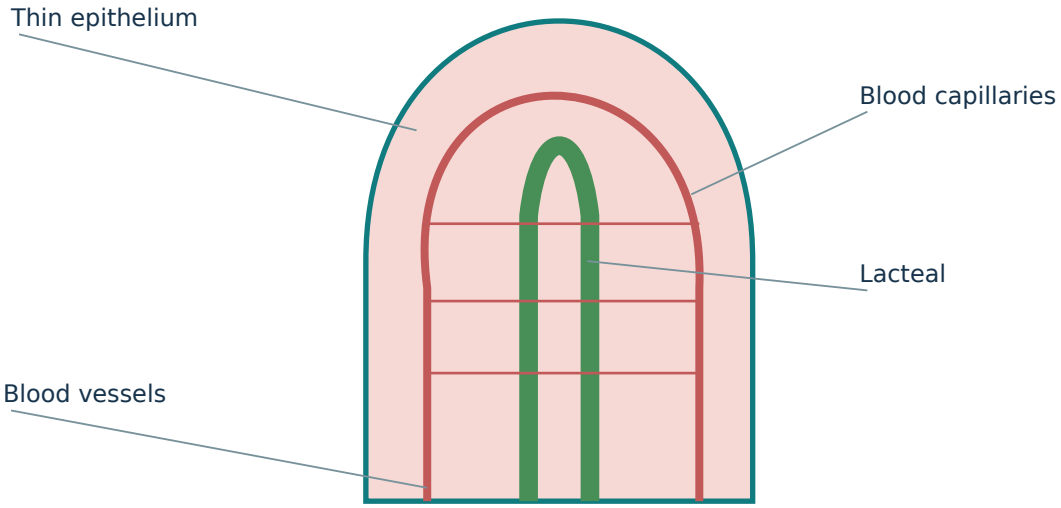
Q.4 | Small intestine: structure and functions [5 marks]

Structure: The small intestine is a long, coiled muscular tube between the stomach and large intestine. It has three regions: duodenum, jejunum and ileum. Its inner wall has folds bearing many villi; epithelial cells have microvilli. Each villus contains blood capillaries and a lacteal.

• Digestion: In the duodenum, bile and pancreatic juice mix with chyme. Bile emulsifies fats and helps reduce acidity. Pancreatic enzymes digest carbohydrates, proteins and fats; enzymes on the intestinal lining complete digestion.

• Absorption: Glucose and amino acids enter blood capillaries. Most absorbed fats enter lacteals after processing in intestinal cells. Water, vitamins and mineral ions are also absorbed.

• Movement: Segmentation mixes food with digestive juices and brings it against the wall. Peristalsis moves the contents towards the large intestine.



Finger-like villus; microvilli add more surface area

Q.4 OR | Four adverse effects of COVID-19 [5 marks]

COVID-19 is caused by the coronavirus SARS-CoV-2. Its severity varies; some people have mild illness while others develop serious disease.

• Respiratory damage: Inflammation of the airways and lungs may cause pneumonia, breathlessness and low blood oxygen.

• General illness: Fever, tiredness, headache and body aches reduce normal activity and may lead to dehydration.

• Circulatory effects: Severe inflammation can damage blood vessels and increase the risk of abnormal clots, sometimes affecting the heart or brain.

• Other organs and recovery: Serious illness may injure organs such as the kidneys. Some patients have lasting tiredness, breathing difficulties or problems with concentration after the acute infection.

Q.5 | Bioremediation and environmental care [5 marks]

Definition: Bioremediation is the use of living organisms, mainly microorganisms or plants, to remove or reduce environmental pollutants.

• Oil spills: Selected bacteria break down hydrocarbons in oil into less harmful substances.

• Sewage treatment: Microorganisms digest organic wastes, reducing pollution before water is released.

• Polluted soil: Microbes break down some pesticides and other organic chemicals.

• Heavy metals: Some plants and microbes absorb or immobilise metals, so they can be removed or made less available. Metals are not destroyed.

This process can clean soil and water with less disturbance and lower cost than some physical or chemical methods. For example, oil-degrading bacteria can help clean an oil-contaminated site.

Q.5 OR | Five uses of biostatistics [5 marks]

Biostatistics applies statistical methods to biological and health data. Five uses are:

• Disease monitoring: Count cases and compare disease rates in different places or time periods.

• Testing treatments: Compare results in groups to assess the effectiveness and safety of medicines or vaccines.

• Health planning: Use patient data to plan beds, staff, medicines and health budgets.

• Risk factors: Study links between factors such as smoking and disease.

• Agriculture and research: Compare crop yields, growth or experimental results to reach evidence-based conclusions.

Q.6 | Structure of the human heart [5 marks]

The heart is a hollow muscular organ in the chest between the lungs. It is enclosed in a protective pericardium. Its wall contains cardiac muscle, called the myocardium.

• Four chambers: The upper right and left atria receive blood. The lower right and left ventricles pump blood out. The left ventricular wall is thickest because it pumps to the whole body.

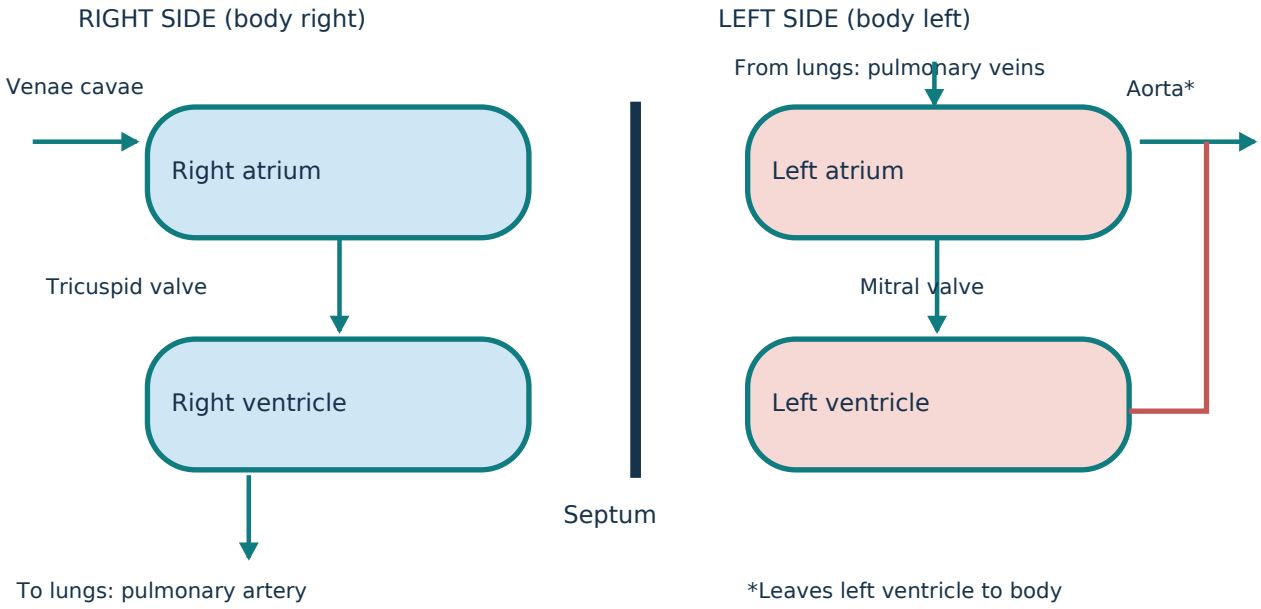
• Septum: Separates the right and left sides and prevents direct mixing of their blood.

• Valves: The tricuspid valve lies between right atrium and right ventricle; the bicuspid/mitral valve lies between left atrium and left ventricle. Semilunar valves lie at the pulmonary artery and aorta. They prevent backflow.

• Major vessels and route: Venae cavae → right atrium → right ventricle → pulmonary artery → lungs → pulmonary veins → left atrium → left ventricle → aorta → body.

• Own blood supply: Coronary arteries supply the heart muscle with oxygen and nutrients.

Heart schematic: arrows show direction of blood flow.

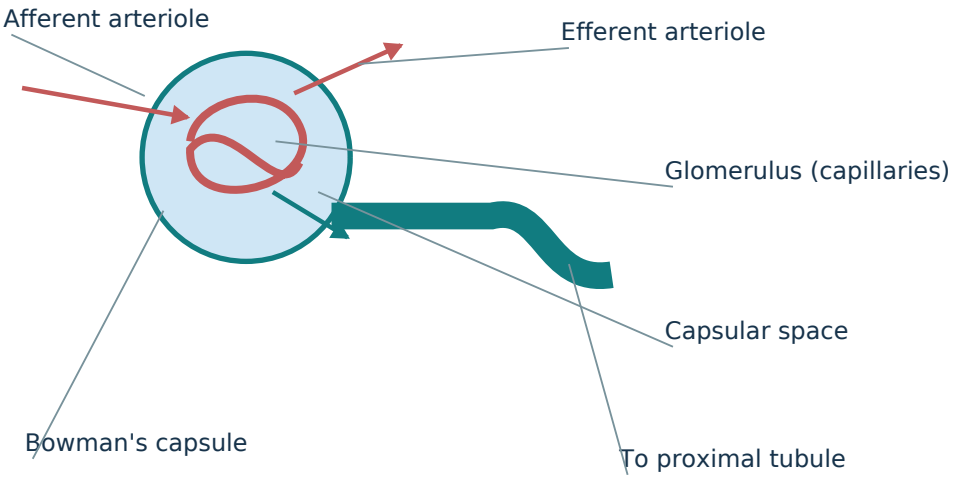


Q.6 OR | Bowman's capsule and glomerulus [5 marks]

Bowman's capsule: A double-walled, cup-shaped structure at the beginning of a nephron in the kidney cortex. Its inner layer has specialised cells called podocytes, with filtration slits. The space between its walls collects filtrate and opens into the proximal tubule.

Glomerulus: A tuft of capillaries inside the capsule. Blood enters through the afferent arteriole and leaves through the efferent arteriole. The capillary walls are adapted for filtration, and the blood pressure is high enough to drive filtration.

Combined function: Water and small solutes, including glucose, amino acids, salts and urea, pass through the filtration barrier into the capsule. Blood cells and most proteins remain in circulation. The filtrate then enters the renal tubule for selective reabsorption and further processing into urine.



Pressure filters water and small solutes into the capsule.

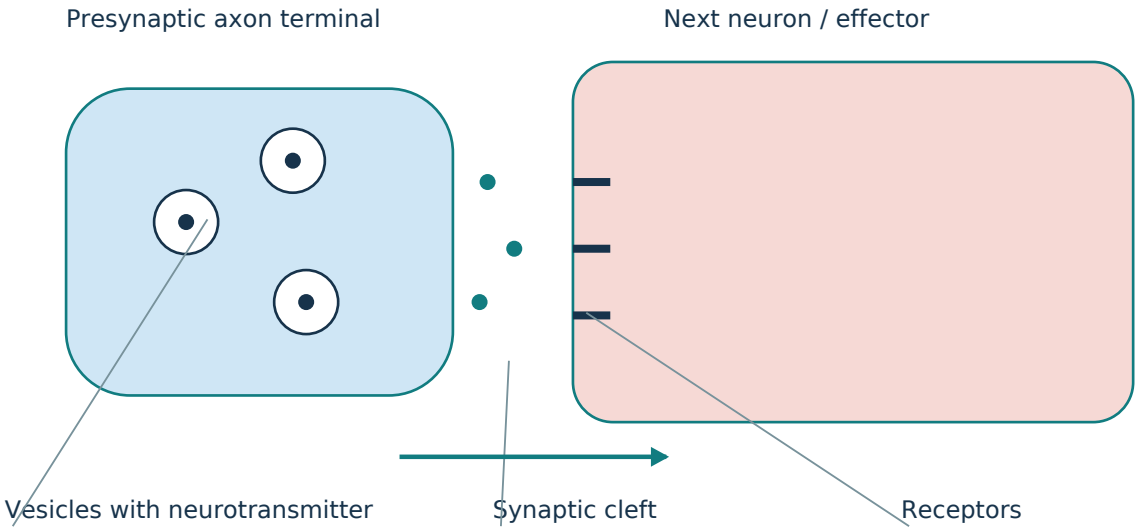
Paper 4: 1st Annual 2026 - Variant M

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Section B: Q.2 (33 marks) - Both choices of every part are answered below for practice. In the examination, follow the choice instructions printed on your paper.

Q.2(i) | Synapse: definition and labelled sketch [3 marks]

A synapse is a junction where a neuron communicates with another neuron or an effector. In a chemical synapse, the cells are separated by a tiny synaptic cleft. An arriving impulse causes vesicles to release a neurotransmitter. It diffuses across the cleft and binds to receptors on the next cell, changing its activity.



Q.2(i) OR | Labels on the neuron [3 marks]

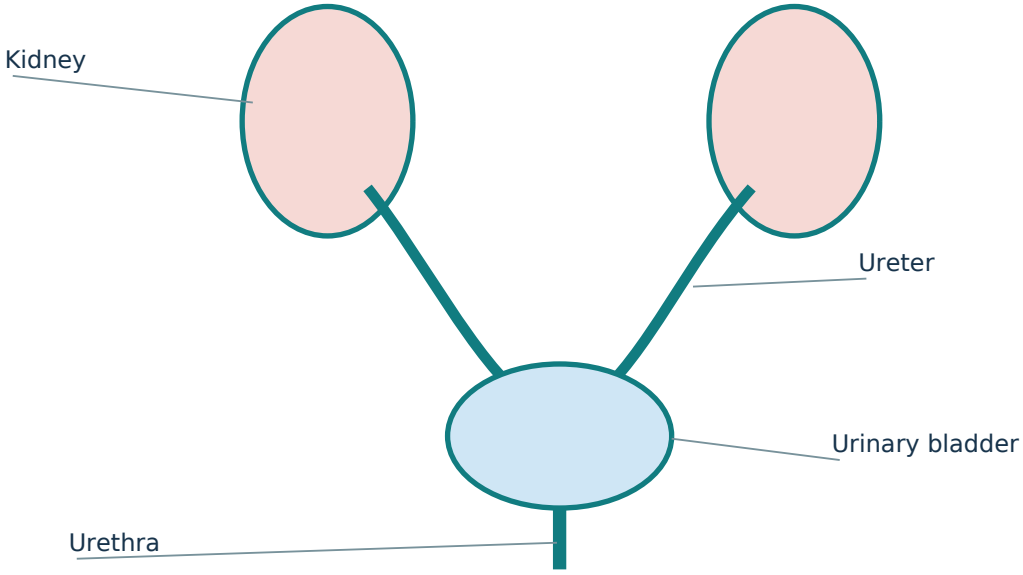
Label	Part	Identification
A	Dendrite	A short, branching process that receives signals towards the cell body.
B	Axon	The long process that carries impulses away from the cell body; the scan shows myelin around it.
C	Cell body / cyton	The region containing the nucleus and most cell organelles.

Q.2(ii) | Parthenogenesis with an example [3 marks]

Parthenogenesis is the development of a new individual from an unfertilised egg. There is no fusion of sperm and egg. For example, in honeybees, unfertilised eggs develop into haploid male drones, while fertilised eggs develop into diploid females.

Q.2(ii) OR | Labelled urinary system [3 marks]

The human urinary system consists of two kidneys, two ureters, a urinary bladder and a urethra. Kidneys form urine; ureters carry it to the bladder, which stores it. The urethra carries urine outside the body.



Q.2(iii) | Why a dihybrid cross gives more variation [3 marks]

A monohybrid cross follows one pair of contrasting traits, while a dihybrid cross follows two. A double-heterozygous parent, such as RrYy, can form four gamete types (RY, Ry, rY and ry) by independent assortment; a single heterozygote, such as Rr, forms only two (R and r). Thus the usual dihybrid F_2 has four phenotypic classes (9:3:3:1), compared with two (3:1) in a monohybrid cross, assuming complete dominance and independent genes.

Q.2(iii) OR | Which air is warmer in winter? [3 marks]

Expired air is usually warmer than inspired air in winter. Cold incoming air is warmed as it passes over the moist lining and rich blood supply of the nasal cavity and other air passages. It also exchanges heat with the warm lungs, so air breathed out is closer to body temperature.

Q.2(iv) | Three agricultural benefits of food biotechnology [3 marks]

• Greater protection: Pest- and disease-resistant crop varieties reduce damage and help protect yield.

• Better planting material: Tissue culture produces many healthy, uniform plants and can supply disease-free stocks.

• Improved food quality: Breeding and genetic methods can improve nutritional content, storage life or tolerance to environmental stress.

Q.2(iv) OR | Why the liver is an important metabolic organ [3 marks]

• Carbohydrate metabolism: It stores extra glucose as glycogen and releases or makes glucose when needed, helping control blood glucose.

• Protein and fat metabolism: It removes amino groups from excess amino acids and converts ammonia into urea. It also processes fats and makes important blood proteins, linking the metabolism of different nutrients.

Q.2(v) | Pulmonary and systemic circulation [3 marks]

Feature	Pulmonary circuit	Systemic circuit
Route	Right ventricle → pulmonary artery → lung capillaries → pulmonary veins → left atrium.	Left ventricle → aorta → body capillaries → venae cavae → right atrium.
Main function	Oxygenates blood and removes carbon dioxide at the lungs.	Supplies tissues with oxygen and nutrients and collects wastes.
Pressure	Relatively low, protecting delicate lung capillaries.	Relatively high to carry blood throughout the body.

Q.2(v) OR | Arteries and veins: three differences [3 marks]

Feature	Arteries	Veins
Direction	Carry blood away from the heart.	Carry blood towards the heart.
Wall and lumen	Thick, muscular, elastic walls; relatively narrow lumen.	Thinner walls with less muscle and elastic tissue; wider lumen.
Pressure and valves	Higher pressure; no valves along their length.	Lower pressure; many have valves to prevent backflow.

Q.2(vi) | Asthma: causes, symptoms and treatment [3 marks]

Cause / trigger: In susceptible people, allergens such as pollen, dust mites or animal dander, and irritants such as smoke, trigger airway inflammation, muscle narrowing and excess mucus.
Symptoms: Wheezing, cough, chest tightness and difficulty breathing.
Treatment: Avoid known triggers. Prescribed inhaled bronchodilators open the airways, while inhaled corticosteroids help control inflammation. Treatment is adjusted by a healthcare professional.

Q.2(vi) OR | Thalamus and hypothalamus [3 marks]

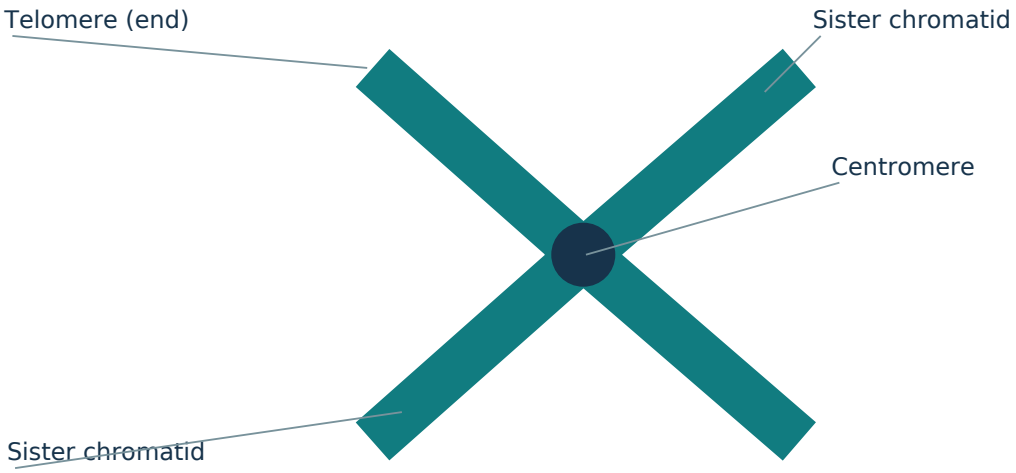
Feature	Thalamus	Hypothalamus
Location	In the forebrain, above the hypothalamus and on either side of the third ventricle.	Below the thalamus, close to and connected with the pituitary gland.
Main function	Relays most sensory information to the cerebral cortex (smell is the major exception).	Maintains homeostasis: controls temperature, hunger, thirst and water balance; regulates autonomic activity and the pituitary.

Q.2(vii) | Spermatogenesis versus oogenesis [3 marks]

Feature	Spermatogenesis	Oogenesis
Site	Seminiferous tubules of testes.	Ovaries.
Products of one primary cell	Four small haploid sperm, usually all functional.	One large haploid ovum and small polar bodies due to unequal cytoplasmic division.
Timing in humans	Starts at puberty and continues through adult life.	Starts before birth; later development is cyclic after puberty, and meiosis II normally finishes only after fertilisation.

Q.2(vii) OR | Chromosome labels P, Q and R [3 marks]

Label	Part	Brief description
P	Telomere	Protective end region of the chromosome.
Q	Centromere	Constricted region joining the sister chromatids; spindle attachment occurs at kinetochores here.
R	Chromatid	One of the two identical halves of a duplicated chromosome; it contains coiled DNA and proteins (chromatin).



Duplicated chromosome: two sister chromatids

Q.2(viii) | One disease transmitted from animals [3 marks]

Rabies is a zoonotic viral disease, meaning it can pass from animals to humans. The virus is usually transmitted through saliva during the bite of an infected dog or another mammal. It attacks the nervous system and may cause fever, agitation, difficulty swallowing and hydrophobia. It is almost always fatal after symptoms begin; prompt medical post-exposure vaccination can prevent it before symptoms develop.

Q.2(viii) OR | Why blood clotting is important [3 marks]

• Prevents blood loss: A clot seals a damaged vessel, helping maintain blood volume and normal blood pressure.

• Protects the wound: The clot forms a temporary barrier that reduces entry of microbes and provides a framework while tissues repair.

Q.2(ix) | Three applications of marine biotechnology [3 marks]

• Food: Aquaculture and improved breeding increase the production of fish and shellfish; algae can provide food and useful nutrients.

• Medicine: Marine organisms provide compounds used or studied for medicines, such as anticancer and antimicrobial drugs.

• Environment and industry: Marine microbes can help break down oil; marine enzymes and algal products are useful in industrial processes.

Q.2(ix) OR | Mode and mean [3 marks]

Data: 3, 5, 6, 6, 9, 10, 9, 6, 6.
Mode = 6, because it occurs four times, more often than any other value.
Sum = 3 + 5 + 6 + 6 + 9 + 10 + 9 + 6 + 6 = 60.
Number of observations = 9.
Mean = 60 ÷ 9 = 6.67 (approximately).

Q.2(x) | How villus structure aids absorption [3 marks]

• Large surface area: Villi and their microvilli provide more area for absorbing digested nutrients.

• Thin wall: The epithelium is one cell thick, so nutrients cross a short distance.

• Good transport: Blood capillaries rapidly carry away sugars and amino acids; the central lacteal takes up most fats. This maintains absorption gradients.

Q.2(x) OR | Angina [3 marks]

Angina is chest pain or discomfort caused when the heart muscle temporarily receives too little oxygen-rich blood, commonly because coronary arteries are narrowed. It often appears during exertion or emotional stress and may spread to the arm, shoulder, neck or jaw. Unlike a myocardial infarction, a short angina episode does not itself mean that heart muscle has died.

Q.2(xi) | Why penicillin was a medical milestone [3 marks]

• Discovery: In 1928, Alexander Fleming noticed that the mould Penicillium notatum prevented bacteria from growing around it.

• Antibiotic: The mould produced penicillin, which stops susceptible bacteria from making proper cell walls.

• Importance: Penicillin made many previously dangerous bacterial infections treatable, saved lives and encouraged the development of other antibiotics. It does not treat viral infections.

Q.2(xi) OR | Dominant and recessive alleles [3 marks]

Dominant: An allele whose effect is expressed in a heterozygous individual. For example, T produces tall pea plants in both TT and Tt.
Recessive: An allele whose effect is masked by a dominant allele in a heterozygote and appears when no dominant allele is present. For example, dwarfness appears in tt pea plants. Dominant does not mean more common or better.

Paper 4: 1st Annual 2026 - Variant M - Section C

Q.3 to Q.6 carry 5 marks each. Both alternatives are answered. Use the labelled diagrams with the written explanation.

Q.3 | Adrenal glands: hormones and functions [5 marks]

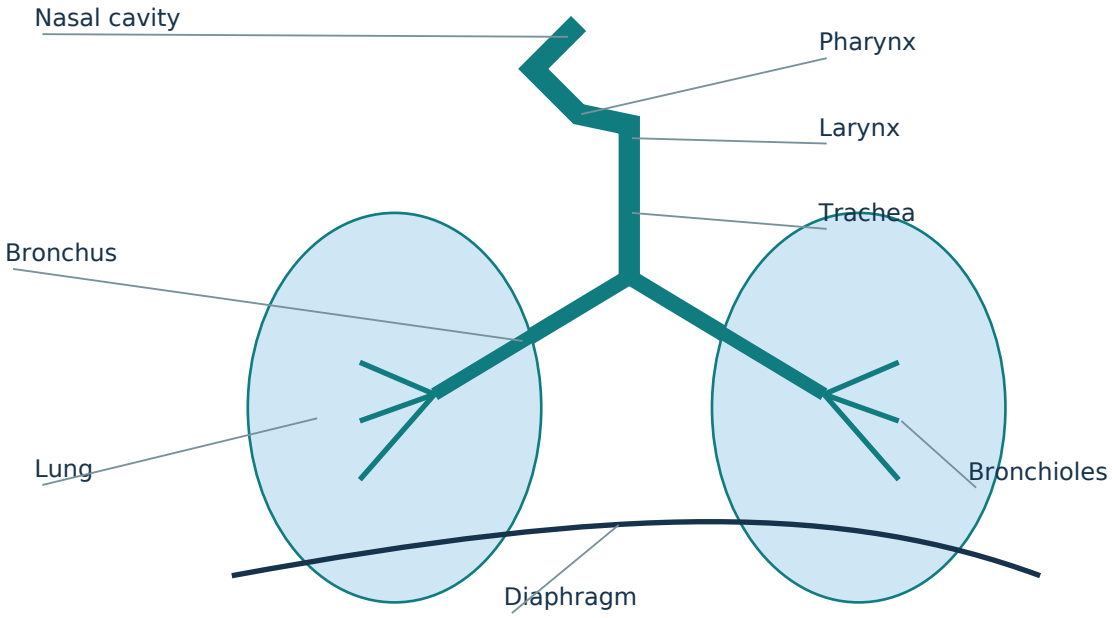
The adrenal glands are a pair of endocrine glands, one on top of each kidney. Each has an outer cortex and an inner medulla.

Region	Hormones	Functions
Adrenal medulla	Adrenaline (epinephrine) and noradrenaline	Prepare the body for fight or flight: increase heart rate and blood delivery to muscles, widen airways and increase available blood glucose.
Adrenal cortex	Cortisol (a glucocorticoid)	Helps the body respond to stress, regulates nutrient metabolism and helps raise blood glucose when required.
Adrenal cortex	Aldosterone (a mineralocorticoid)	Increases sodium reabsorption and potassium secretion by the kidneys; water follows sodium, helping maintain blood volume and pressure.
Adrenal cortex	Small amounts of adrenal androgens	Contribute to development of some sexual characteristics, including body hair.

Overall role: These hormones support the stress response, metabolism and normal salt-water balance.

Q.3 OR | Air passageway and roles of two parts [5 marks]

Pathway: Nostrils → nasal cavity → pharynx → larynx → trachea → right and left bronchi → bronchioles → alveoli.



• Nasal cavity: Its hairs and mucus trap dust and microbes. Its moist lining adds water vapour and its rich blood supply warms incoming air.

• Trachea: Carries air towards the bronchi. Cartilage rings prevent collapse; mucus traps particles and cilia move the mucus towards the throat for removal.

The bronchi and bronchioles distribute air through the lungs. Alveoli are the final gas-exchange surfaces, where oxygen enters blood and carbon dioxide leaves it.

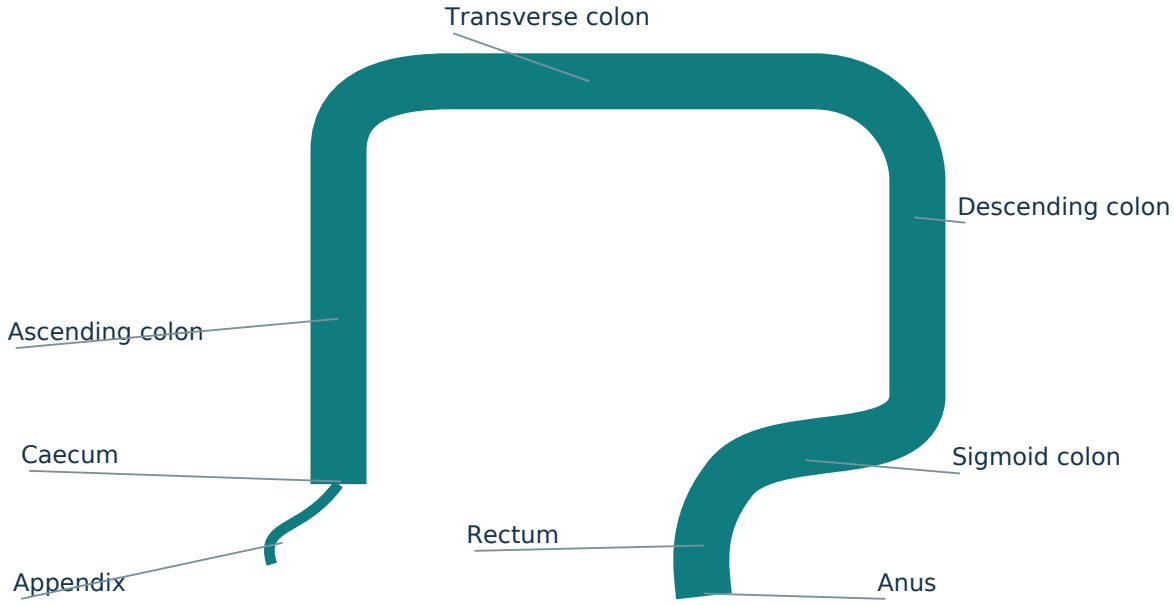
Q.4 | Large intestine: structure and functions [5 marks]

Structure: The large intestine is wider but shorter than the small intestine. It includes the caecum, the colon (ascending, transverse, descending and sigmoid), the rectum and the anal canal. A small appendix projects from the caecum. Unlike the small intestine, its lining has no villi and contains many mucus-secreting cells.

• Water and salt absorption: Absorbs remaining water and some mineral ions from the intestinal contents, helping turn them into faeces.

• Bacterial activity: Normal gut bacteria act on undigested material and make some vitamins, including vitamin K, some of which can be absorbed.

• Storage and egestion: Mucus lubricates the contents. The rectum stores faeces until the defecation reflex expels them through the anus; anal sphincters control release.



Q.4 OR | Plant diseases: rust and smut [5 marks]

A. Rust

• Cause: Rust diseases are caused by fungi, commonly Puccinia species in wheat.

• Symptoms and damage: Rust-coloured, yellow-orange or brown spore pustules develop on leaves and stems. Damage reduces photosynthesis, weakens plants and lowers grain yield.

• Spread and control: Spores spread mainly by wind. Resistant varieties, appropriate crop management and suitable fungicides help control disease. For some rusts, removal of an alternate host also helps.

B. Smut

• Cause: Smut diseases are caused by fungi, such as Ustilago species in cereal crops.

• Symptoms and damage: Grains or other plant parts are replaced by black, powdery masses of fungal spores, reducing crop quality and yield.

• Spread and control: Depending on the species, infection spreads through infected seed, soil or airborne spores. Use certified disease-free seed, resistant varieties and suitable seed treatment; remove infected material where practical.

Q.5 | Bioremediation and environmental care [5 marks]

Definition: Bioremediation is the use of living organisms, mainly microorganisms or plants, to remove or reduce environmental pollutants.

• Oil spills: Selected bacteria break down hydrocarbons in oil into less harmful substances.

• Sewage treatment: Microorganisms digest organic wastes, reducing pollution before water is released.

• Polluted soil: Microbes break down some pesticides and other organic chemicals.

• Heavy metals: Some plants and microbes absorb or immobilise metals, so they can be removed or made less available. Metals are not destroyed.

This process can clean soil and water with less disturbance and lower cost than some physical or chemical methods. For example, oil-degrading bacteria can help clean an oil-contaminated site.

Q.5 OR | Median, mode, mean and worked calculation [5 marks]

Median: The middle value after arranging observations in order; for an even number, average the two middle values.
Mode: The value occurring most frequently.
Mean: The sum of observations divided by their number.

Given data: 4, 6, 8, 10, 12, 6, 8, 9, 8, 4, 12.
Ordered data: 4, 4, 6, 6, 8, 8, 8, 9, 10, 12, 12.
There are 11 values. The middle is the 6th value.
Median = 8.
The value 8 occurs three times, more than any other value.
Mode = 8.
Sum = 4 + 6 + 8 + 10 + 12 + 6 + 8 + 9 + 8 + 4 + 12 = 87.
Mean = 87 ÷ 11 = 7.91 (approximately).

Q.6 | Five essential functions of blood cells [5 marks]

• Oxygen transport: RBC haemoglobin carries oxygen from lungs to tissues, enabling aerobic respiration and energy release.

• Carbon dioxide removal: RBCs help convert CO_2 into bicarbonate and carry some CO_2 on haemoglobin, helping transport it to the lungs.

• Pathogen destruction: Phagocytic WBCs, such as neutrophils and macrophages, engulf harmful microbes and dead cells.

• Specific immunity: B lymphocytes provide antibody-based defence; T lymphocytes coordinate defence or kill infected cells, and memory cells support faster future responses.

• Blood clotting: Platelets help form a plug and initiate clotting, preventing excessive blood loss after injury. Platelets are cell fragments, commonly grouped with the formed elements of blood.

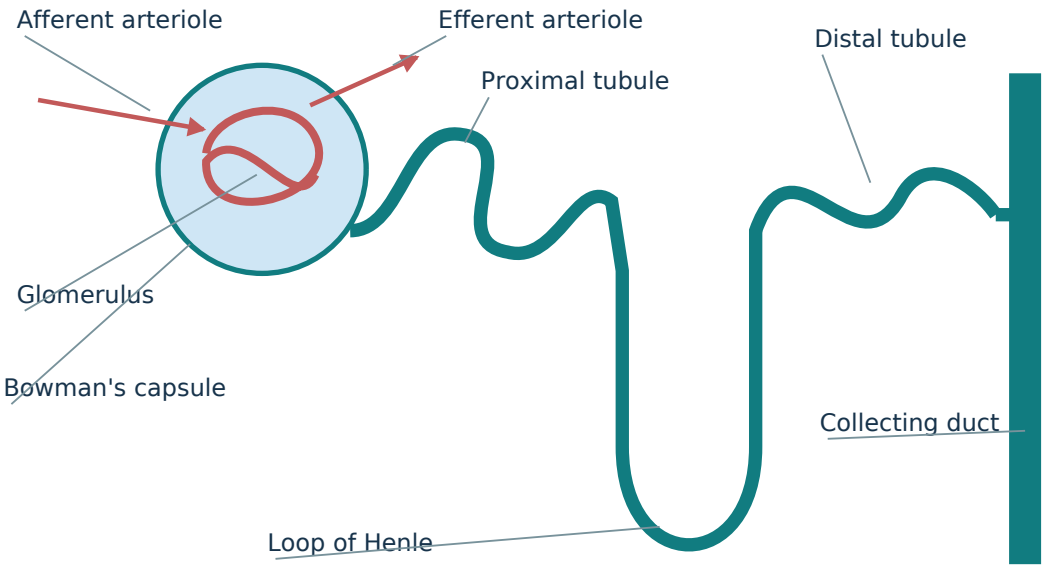
Q.6 OR | Structure of a nephron and labelled diagram [5 marks]

A nephron is the microscopic structural and functional unit of a kidney. Each kidney contains about one million nephrons.

• Renal corpuscle: A cup-shaped, double-walled Bowman's capsule surrounds a network of blood capillaries called the glomerulus. The afferent arteriole brings blood in; the efferent arteriole carries it out.

• Renal tubule: The capsule opens into the proximal convoluted tubule, followed by the descending and ascending limbs of the loop of Henle and the distal convoluted tubule.

• Collecting duct: Receives fluid from several nephrons and carries urine towards the renal pelvis. A capillary network surrounds the tubule. The corpuscle and convoluted tubules lie mainly in the cortex; loops extend into the medulla.



Quick revision: calculations and answer checks

Paper / question	Verified result
Model, Q.6 OR	Mean = 7; mode = 4.
Variant B, Q.2(ix) OR	Mean = 16.67 patients per month; mode = 22 patients.
Variant D, Q.2(vii)	Median = 11 fruits; mean = 11.2 fruits per tree.
Variant M, Q.2(ix) OR	Mode = 6; mean = 6.67.
Variant M, Q.5 OR	Mean = 7.91; median = 8; mode = 8.

Common mistakes to avoid

• The subjective total is 53: Section B = 33 and Section C = 20.

• Artery / vein describes direction relative to the heart, not oxygen content. Pulmonary arteries carry deoxygenated blood; pulmonary veins carry oxygenated blood.

• Bile emulsifies fat; it contains no digestive enzyme. Pepsin digests proteins, not starch.

• The liver forms urea; the kidneys remove it from blood.

• Filtration moves small substances from glomerular blood into the capsule; reabsorption returns useful substances from tubules to blood.

• Angina is temporary inadequate oxygen supply; a myocardial infarction involves death of heart muscle.

• Most carbon dioxide travels as bicarbonate, not attached to haemoglobin.

• Mean requires every value; median requires sorting; mode requires counting frequencies.

Coverage: Four papers × [22 Section-B answers + 8 Section-C answers] = 120 answered choices. Diagrams and repeated topics are included where they help each paper stand on its own.

Source and alignment notes

Question wording, order, paper codes and marks were checked against the user-supplied five-page PDF, FBISE_Biology_SSC_II_Past_Papers_and_Model_Paper_Sec_B_and_C.pdf. Short question titles in this solution are paraphrases; no questions or OR choices have been intentionally omitted.

FBISE states that its examinations follow the relevant curriculum SLOs and assessment frameworks rather than one textbook. The answers therefore emphasise the biological concept, the stated command (define, compare, explain, calculate or draw) and each requested subpart. No official 2026 examiner-specific rubric was supplied; this document does not claim official endorsement.

Official reference pages:
1. FBISE assessment frameworks and model papers:
https://www.fbise.edu.pk/curriculum_model_paper.php
2. FBISE SSC-II old question papers:
https://www.fbise.edu.pk/AllOldPapersSSC2.php
3. FBISE notification: Implementation of Assessment Frameworks (Curriculum 2022-23), available through the framework page above.